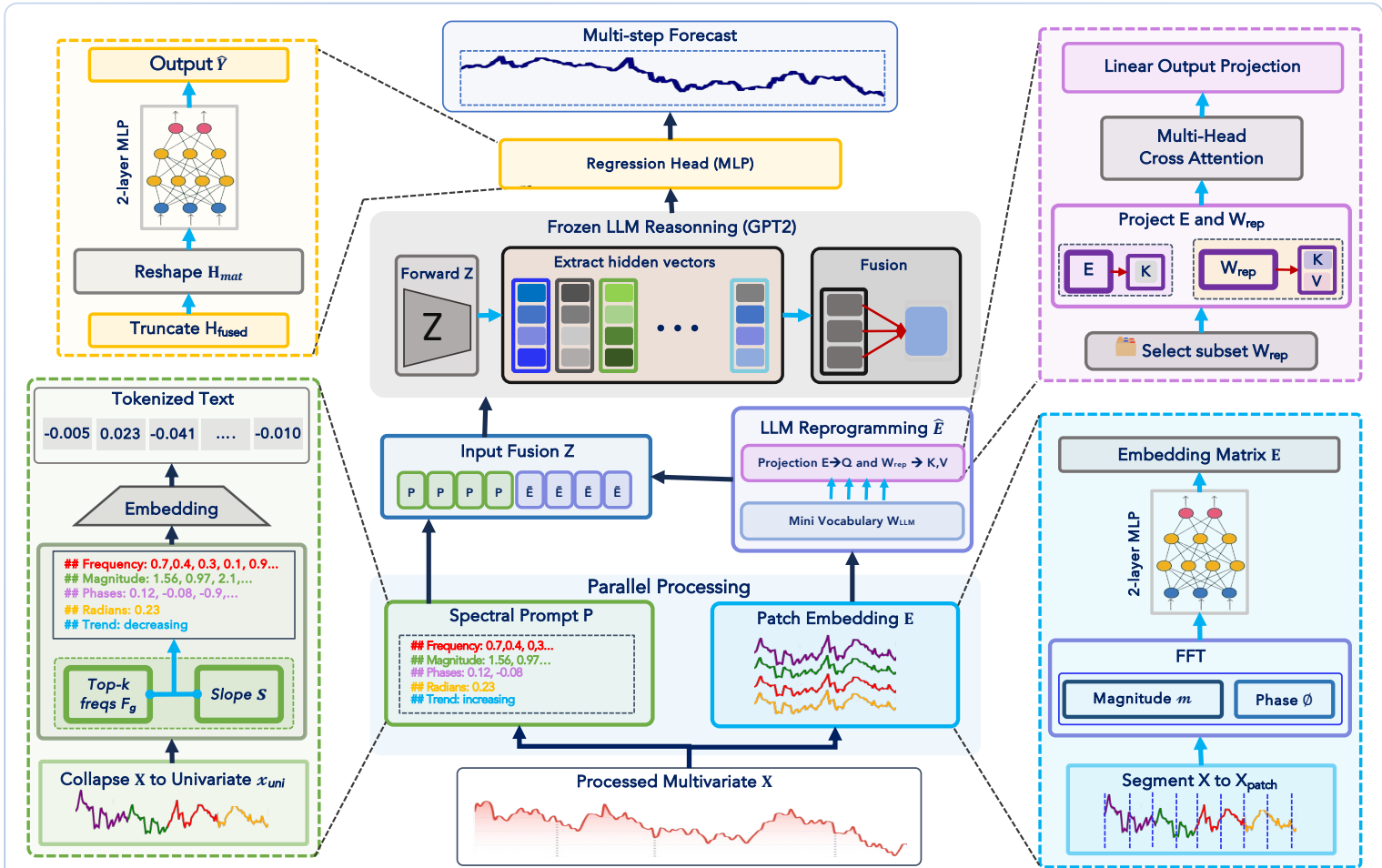

X : Multivariate Input Data

$\mathbf{E}$: Patch Embedding

$\mathbf{P}$: Spectral Prompt

$\hat{E}$: Reprogrammed Tokens

$\mathbf{Z}$: Fused $\mathbf{P}$ and $\hat{E}$

$\hat{Y}$: Predicted Output

m : Magnitude

$\emptyset$: Phase

FFT : Fast Fourrier Transform

x_{uni} : Univariate Signal

F_g : Dominant Frequencies

s : Slote to direct the trend

H_{fused} : Condensed Global Representation

H_{mat}: Token-Structured Forecasting Input

Q, K, V: Query — Key — Value

W_{rep}: Reprogramming Tokens

W_{LLM}: Compact Mini Vocabulary

MLP : Multi-layer Perceptron